

Module 02: Network Topologies & Design Trade-offs

AKTU BCS603 Focus: Architectural layout, link calculation formulas, fault isolation, and comparative evaluation of Mesh, Star, Bus, Ring, and Hybrid topologies.

Physical Network Topologies: Geometric Layouts & Formulas

Mesh Topology

Links = $n(n-1)/2$
Ports/Node = $(n - 1)$

Star Topology

Links = n | Ports = 1
Central Single Point of Failure

Bus Topology

1 Backbone + n Drop Lines
Multipoint Shared Medium

Ring Topology

Links = n | Ports = 2
Token Ring (Unidirectional)

Topology	Physical Links	I/O Ports/Node	Cable Cost	Fault Tolerance	Key Bottleneck / Failure Point
Mesh	$n(n - 1) / 2$	$n - 1$	Very High	Maximum (Robust)	Excessive cabling & port requirement
Star	n	1	Moderate	Good (1 node fail ok)	Central Hub Failure collapses entire network
Bus	1 backbone + n drops	1 tap	Lowest	Poor	Backbone cable break stops all transmission
Ring	n (unidirectional)	2	Low to Moderate	Low (Dual ring helps)	Single break halts ring; token circulation delay
Tree/Hybrid	Hierarchical / Varies	Varies	Moderate to High	Isolated to segment	Root switch or trunk link failure isolates branch

1. Mathematical Formulation Summary (\$n\$ Nodes)

Topology	Link Formula	Ports per Device	Cabling Volume	Reliability Profile
Mesh	$n(n - 1) / 2$	$n - 1$	$O(n^2)$ — Maximum	Exceptional; Alternate paths exist for every node pair
Star	n	1	$O(n)$ — Moderate	High node resilience; Vulnerable to Central Hub failure
Bus	1 backbone + n drops	1 tap	$O(1)$ trunk — Lowest	Low; Backbone cut collapses whole segment
Ring	n	2 (In / Out)	$O(n)$ — Low	Moderate; Single node failure breaks token circulation

2. Star vs Mesh Comparison (Exam Favorite)

- Mesh Topology:** Provides guaranteed bandwidth and zero collision, but cable cost scales as $O(n^2)$. Practically deployed in core backbone networks (ISP Core, Defense).
- Star Topology:** Industry standard for Enterprise LANs (10/100/1000Base-T Ethernet). Cheap twisted pair cables, easy troubleshooting, plug-and-play expansion. Central failure risk is mitigated by using dual redundant Managed Switches.

AKTU Numerical Tip: Numerical solve karte waqt hamesha $n(n-1)/2$ formula likhein aur har step me values substitute karein: Agar $n=10$, Links = $(10 \times 9)/2 = 45$ cables!